

Deep Residual Learning for Image Recognition

Kaiming He, Xiangyu Zhang, Shaoqing Ren, Jian Sun

Abstract

On COCO object detection, deep residual nets yield 28% relative improvement solely from extremely deep representations. Faster R-CNN with ResNet-101 outperforms VGG-16 backbone. Submissions won 1st place on ImageNet detection, localization, COCO detection, and COCO segmentation. Deep residual networks represent a fundamental breakthrough in training very deep neural networks. The key insight is that learning residual functions $F(x)=H(x)-x$ with reference to layer inputs is more effective than learning unreferenced functions. When deeper networks converge, a degradation problem emerges where accuracy saturates then degrades rapidly, not caused by overfitting but by optimization difficulties. Training uses SGD with momentum of 0.9 and weight decay of $1e-4$. Learning rate starts at 0.1 and is divided by 10 when error plateaus. Batch size is 256. Data augmentation includes random horizontal flipping, random cropping with padding of 4 pixels, and standard color augmentation as in Krizhevsky et al. The residual learning framework reformulates layers as learning residual functions $F(x)=H(x)-x$. The original mapping becomes $F(x)+x$, realized by feedforward networks with shortcut connections performing identity mapping. These shortcuts add no extra parameters or computational complexity. The entire network trains end-to-end via SGD with backpropagation.

1 Introduction

ResNet uses batch normalization after each convolution before activation. BN addresses internal covariate shift. With BN, plain networks decrease training error with depth but degradation persists. Residual learning with BN achieves consistently lower training error across all depths tested. Training uses SGD with momentum of 0.9 and weight decay of $1e-4$. Learning rate starts at 0.1 and is divided by 10 when error plateaus. Batch size is 256. Data augmentation includes random horizontal flipping, random cropping with padding of 4 pixels, and standard color augmentation as in Krizhevsky et al. The plain network baseline follows VGG design philosophy with mostly 3×3 filters. For same output feature map size, layers have same number of filters. When feature map size is halved, filter count doubles to preserve time complexity. Downsampling is performed by convolutions with stride of 2.

The residual learning framework reformulates layers as learning residual functions $F(x) = H(x) - x$. The original mapping becomes $F(x) + x$, realized by feedforward networks with shortcut connections performing identity mapping. These shortcuts add no extra parameters or computational complexity. The entire network trains end-to-end via SGD with backpropagation. Analysis on CIFAR-10 with 1202-layer networks shows increasing depth can still improve accuracy, though very deep models may overfit. The 110-layer ResNet achieves lower error than its plain counterpart. The bottleneck design uses 1×1 , 3×3 , 1×1 convolutions where 1×1 layers reduce and restore dimensions. The plain network baseline follows VGG design philosophy with mostly 3×3 filters. For same output feature map size, layers have same number of filters. When feature map size is halved, filter count doubles to preserve time complexity. Downsampling is performed by convolutions with stride of 2.

Experiments on ImageNet show ResNet-152 achieves 3.57% top-5 error, 8x deeper than VGG with lower complexity. The 34-layer ResNet achieves lower training error than plain 34-layer network, confirming degradation is addressed. The 152-layer ensemble won 1st place in ILSVRC 2015 classification task. Training uses SGD with momentum of 0.9 and weight decay of $1e-4$. Learning rate starts at 0.1 and is divided by 10 when error plateaus. Batch size is 256. Data augmentation includes random horizontal flipping, random cropping with padding of 4 pixels, and standard color augmentation as in Krizhevsky et al. Analysis on CIFAR-10 with 1202-layer networks shows increasing depth can still improve accuracy, though very deep models may overfit. The 110-layer ResNet achieves lower error than its plain counterpart. The bottleneck design uses 1×1 , 3×3 , 1×1 convolutions where 1×1 layers reduce and restore dimensions.

2 Related Work

Analysis on CIFAR-10 with 1202-layer networks shows increasing depth can still improve accuracy, though very deep models may overfit. The 110-layer ResNet achieves lower error than its plain counterpart. The bottleneck design uses 1x1, 3x3, 1x1 convolutions where 1x1 layers reduce and restore dimensions. The bottleneck building block uses three layers: 1x1 conv reduces dimensions, 3x3 conv performs processing, 1x1 conv restores dimensions. This design reduces computational cost for each residual block. The parameter-free identity shortcuts are particularly important for bottleneck designs. The plain network baseline follows VGG design philosophy with mostly 3x3 filters. For same output feature map size, layers have same number of filters. When feature map size is halved, filter count doubles to preserve time complexity. Downsampling is performed by convolutions with stride of 2.

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3.1 Core Formulation

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3.2 Implementation Details

Experiments on ImageNet show ResNet-152 achieves 3.57% top-5 error, 8x deeper than VGG with lower complexity. The 34-layer ResNet achieves lower training error than plain 34-layer network, confirming degradation is addressed. The 152-layer ensemble won 1st place in ILSVRC 2015 classification task. Analysis on CIFAR-10 with 1202-layer networks shows increasing depth can still improve accuracy, though very deep models may overfit. The 110-layer ResNet achieves lower error than its plain counterpart. The bottleneck design uses 1x1, 3x3, 1x1 convolutions where 1x1 layers reduce and restore dimensions. On COCO object detection, deep residual nets yield 28% relative improvement solely from extremely deep representations. Faster R-CNN with ResNet-101 outperforms VGG-16 backbone. Submissions won 1st place on ImageNet detection, localization, COCO detection, and COCO segmentation.

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Table 1: Benchmark comparison

Method	Dataset	Acc (%)	F1
Baseline	ImageNet	72.3	0.71
ResNet-50	ImageNet	76.1	0.75
ResNet-152	ImageNet	78.3	0.77
Ours	ImageNet	79.1	0.78

4.1 Datasets and Setup

The plain network baseline follows VGG design philosophy with mostly 3x3 filters. For same output feature map size, layers have same number of filters. When feature map size is halved, filter count doubles to preserve time complexity. Downsampling is performed by convolutions with stride of 2. The residual learning framework reformulates layers as learning residual functions $F(x)=H(x)-x$. The original mapping becomes $F(x)+x$, realized by feedforward networks with shortcut connections performing identity mapping. These shortcuts add no extra parameters or computational complexity. The entire network trains end-to-end via SGD with backpropagation. Training uses SGD with momentum of 0.9 and weight decay of $1e-4$. Learning rate starts at 0.1 and is divided by 10 when error plateaus. Batch size is 256. Data augmentation includes random horizontal flipping, random cropping with padding of 4 pixels, and standard color augmentation as in Krizhevsky et al.

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4.2 Main Results

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